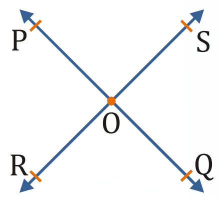
Topper's Toolkit

Lines and Angles

Key Points

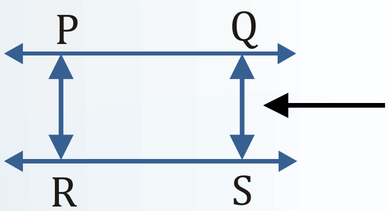
1 Intersecting lines

- Two distinct lines are intersecting, if they have a common point.
- The common point is called the "point of intersection" of the two lines.
- For e.g. PQ and RS are intersecting lines.



2 Non-Intersecting lines (Parallel lines)

- Two distinct lines which are not intersecting are said to be parallel lines.
- The parallel lines are always at a constant distance from each other.
- For e.g. PQ and RS are intersecting lines.

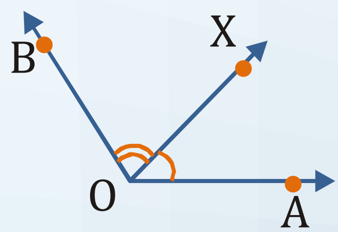


Distance between the lines is always constant

3 Pairs of angles

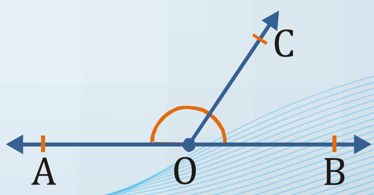
Adjacent angles

- Two angles are called adjacent angles, if :
  (i) They have the same vertex.
  (ii) They have a common arm, and
  (iii) Uncommon arms are on either side of the common arm.



Linear pair of angles

- Two adjacent angles are said to form a linear pair of angles if non-common arms are two opposite rays.



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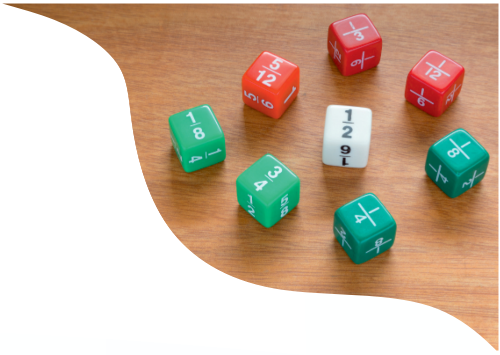
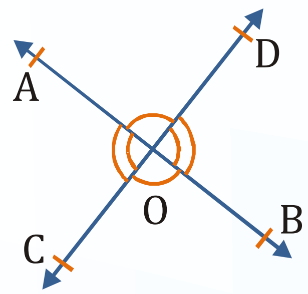
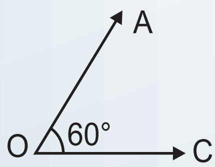


Vertically opposite angles

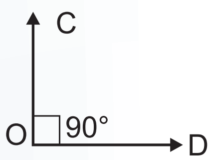
- If two lines intersect each other, then the pairs of opposite angles formed are called vertically opposite angles.
- For e.g. $\angle AOC = \angle BOD$ and $\angle AOD = \angle BOC$



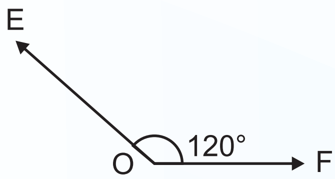
Type of Angles



Acute Angle
Less than 90°



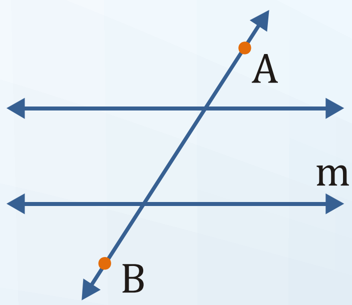
Right Angle
Equal to 90°

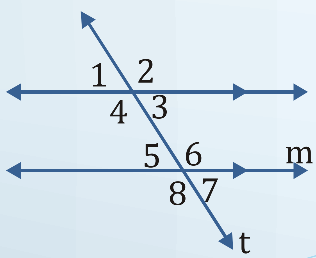


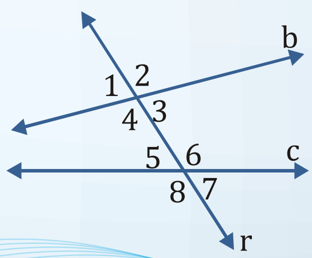
Obtuse Angle
Between 90° and 180°.

Parallel lines and transversals

- A line, line segment or ray that intersects two or more lines at different points is called a transversal.
- For e.g. $\overleftrightarrow{AB}$ is a transversal. It intersects lines ℓ and m.
- The lines cut by a transversal may or may not be parallel.
- When a transversal intersects two lines, eight angles are formed.

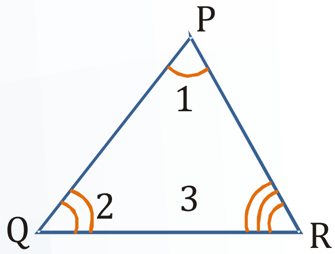






Angle sum property of a triangle

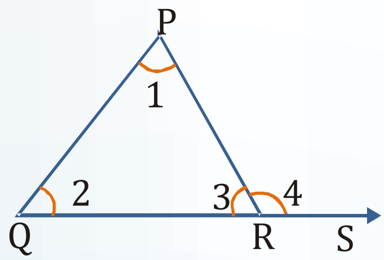
Theorem : The sum of all the angles of a triangle is 180°.



In a Δ PQR, $\angle 1 + \angle 2 + \angle 3 = 180°$.

Exterior angle property of a triangle

Theorem : If a side of a triangle is produced, then the exterior angle so formed is equal to the sum of the two interior opposite angles.



In a ΔPQR, $\angle 4 = \angle 1 + \angle 2$.